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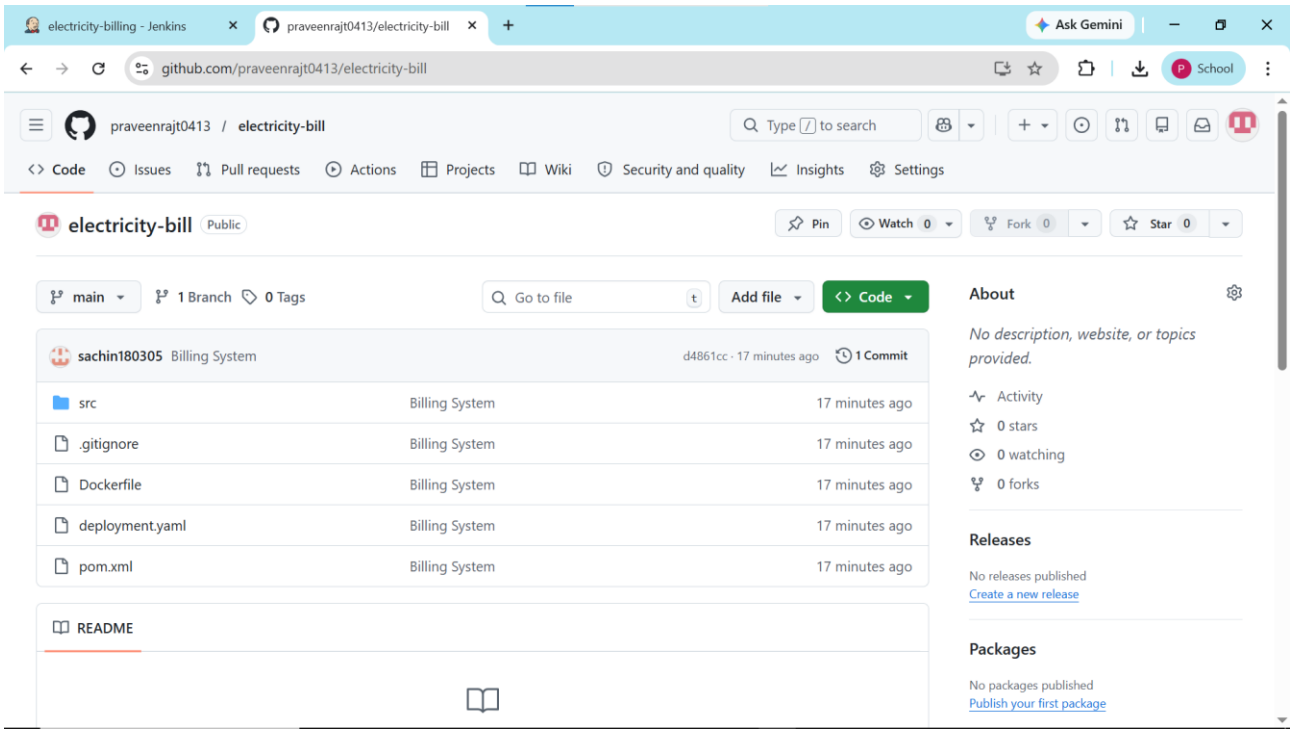
Agile Lab Fat
SET 6
ANSWER ALL THE QUESTIONS
Smart Electricity Billing System
Problem Statement: Ensure electricity usage is calculated and billed correctly.
Tasks:
1. Develop billing calculation logic
2. Write JUnit tests for tariff calculations
3. Implement CI/CD pipeline with automated testing
➢ Set up version control using GitHub
➢ Configure CI pipeline (e.g., Jenkins/GitHub Actions) to trigger on code commits
➢ Automate build process using Maven
➢ Integrate automated unit tests in pipeline
➢ Configure CD pipeline for automatic deployment to staging/production
4. Deploy services on Docker and Kubernetes
➢ Containerize application using Docker (write Dockerfile)
➢ Build and push Docker image to container registry (Docker Hub)
➢ Create Kubernetes deployment and service YAML files

Files in github:
https://github.com/praveenrajt0413/electricity-bill.git



Junit or Maven Testing:

Jenkins CI/CD Pipeline:

Problems Output Debug Console Terminal Ports powershell

```
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.302 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ simplejava ---
[INFO] Building jar: C:\Users\user\OneDrive\Desktop\praveen1\target\simplejava.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  16.393 s
[INFO] Finished at: 2026-04-15T18:33:42+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\user\OneDrive\Desktop\praveen1>
```

#1 - electricity-billing - Jenkins | praveenrajt0413/electricity-bill | Ask Gemini

localhost:8080/job/electricity-billing/1/pipeline-overview/

Jenkins / electricity-billing / #1 / Pipeline Overview

#1

Rerun

Manually run by Praveen Raj | Started 9.6 sec ago | Queued 52 ms | Took 9.5 sec and counting

Graph

Start → Clone Repository → Build & Run Tests → Package Application → Build Docker Image → Push to DockerHub → Deploy to Kubernetes → End

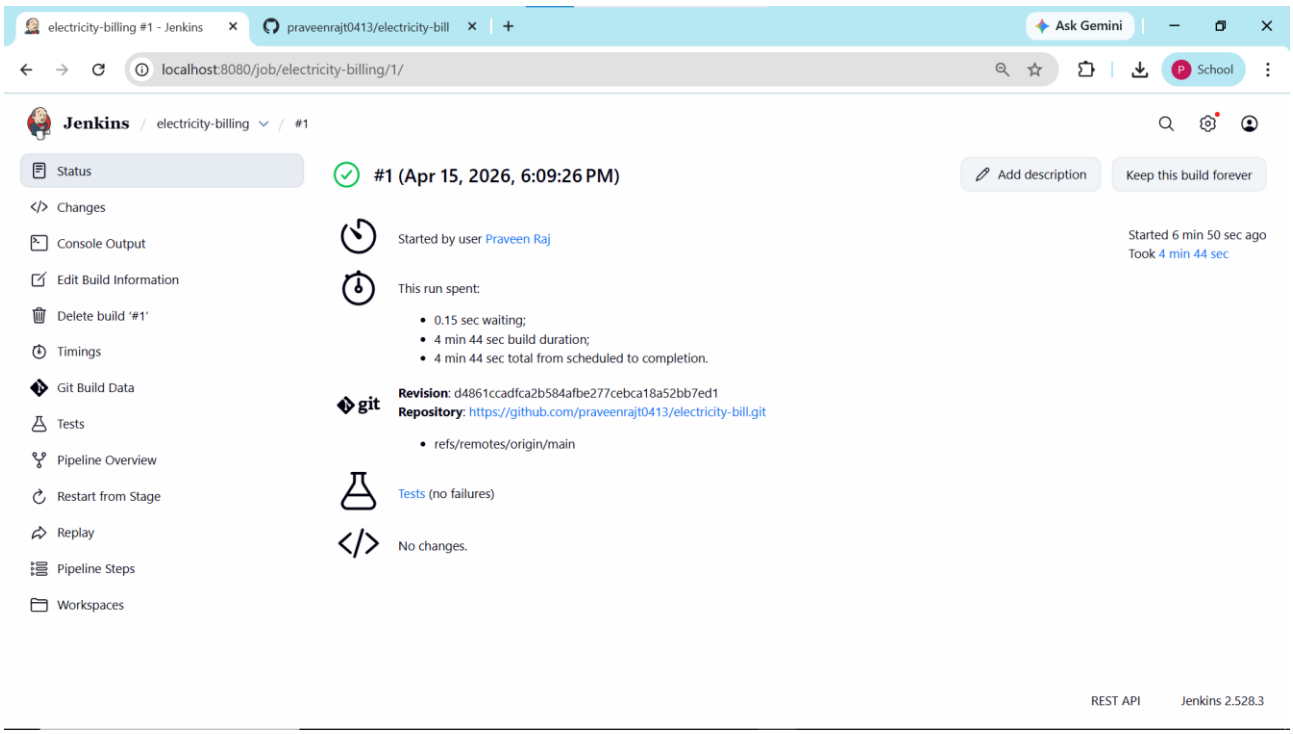
Search

- Clone Repository 5.2s
- Build & Run Tests 51s
- Package Application 39s
- Build Docker Image 2m 43s
- Push to DockerHub 16s

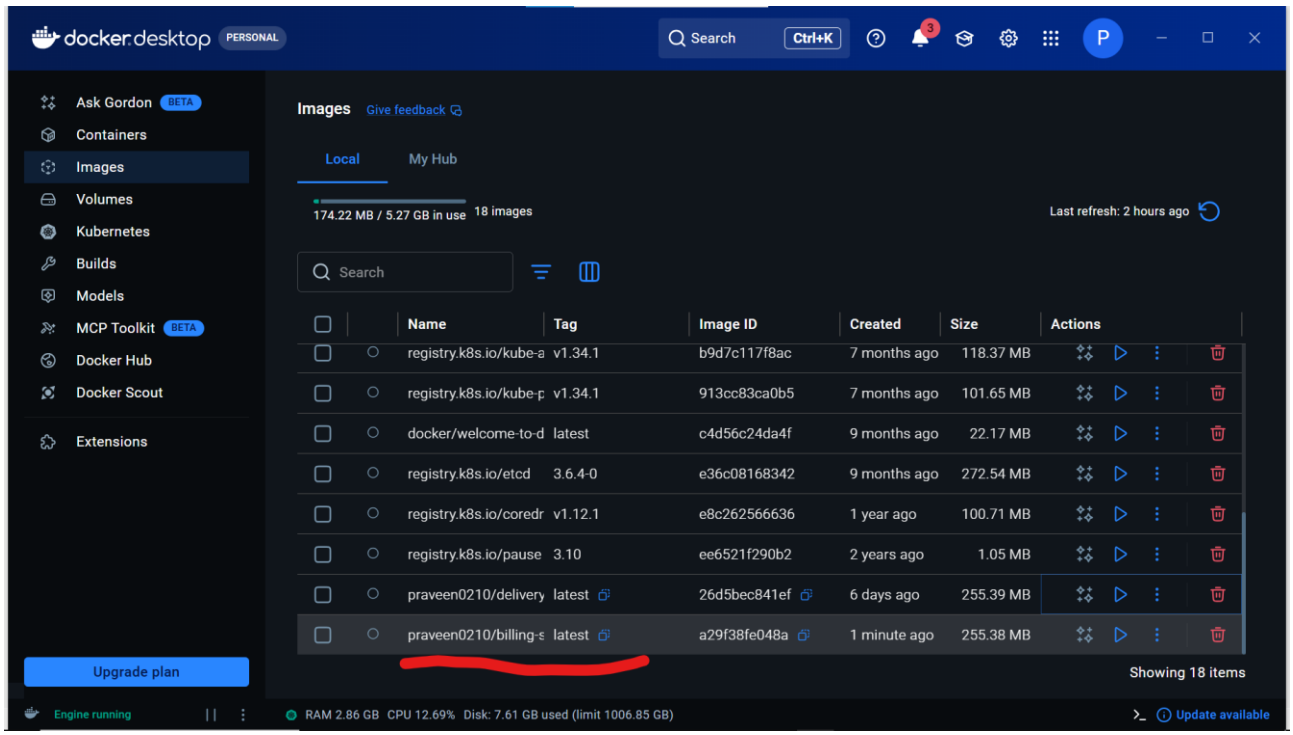
Deploy to Kubernetes 4.0s Started 7.6s ago Jenkins

Windows Batch Script `kubectl apply -f deployment.yaml` 3.5s

```
0
1 C:\ProgramData\Jenkins\.jenkins\workspace\electricity-billing>kubectl apply -f deployment.yaml
2 deployment.apps/delivery-system-deployment unchanged
3 service/delivery-system-service unchanged
```



Docker:



Kubernetees:

Command Prompt

```
Microsoft Windows [Version 10.0.19045.7184]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>cd C:\Users\user\OneDrive\Desktop\praveen1

C:\Users\user\OneDrive\Desktop\praveen1>kubectl get svc
NAME                      TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
delivery-system-service   NodePort    10.104.127.198   <none>        80:30007/TCP   6d5h
kubernetes                ClusterIP   10.96.0.1        <none>        443/TCP        6d6h

C:\Users\user\OneDrive\Desktop\praveen1>
```